

SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-18 05:16 UTC

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Generated 2026-05-18 05:16 UTC

- Cycles recorded: **767**
- Time span: 583.6h (~1.31 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp_notebook.jsonl

Verdict distribution

Verdict	Count
INCONCLUSIVE	727
BARRIER_HIT	20
FALSIFIED	16
SUPPORTED	4

Mathematical fields explored (field_A)

Field	Cycles
Real Algebraic Geometry	40
Schur-Weyl Duality	28
Free Probability Theory	19
Representation Theory of Symmetric Groups	18
Matroid Theory	16
Free Probability	14
Additive Combinatorics	13
Noncommutative L^p Geometry	10
Random Matrix Theory	8
Polymatroid Theory	8
Algebraic Geometry	7
Spectral Graph Theory	7
Finite Geometry	6
Plethysm Theory	6
Persistent Homology	5
Noncommutative Harmonic Analysis	5
Ergodic Circuit Framework (communication complexity via dynamical systems)	4
{“framework_name”: “Ergodic Circuit Framework”, “math_branch”: “COMM_COMPLEXITY”}	4
Fourier Analysis on Boolean Functions	4

Field	Cycles
Algebraic Geometry over Finite Fields	4
Fourier Analysis of Boolean Functions	4
Algebraic Geometry of Secant Varieties	4
FOURIER_ANALYSIS	4
Noncommutative Geometry	4
Fourier analysis of boolean functions	3
Algebraic Topology	3
matroid theory	3
Coarse Geometric Lifting (CGL) — Geometric Complexity Theory	3
{“framework_name”: “Tropical Circuit Weight Analysis (TCWA)”, “math_branch”: “BOUNDED_ARITHMETIC”}	3
Tropical Circuit Weight Analysis (BOUNDED_ARITHMETIC)	3

Last 15 cycles (chronological)

Time	Verdict	Title
2026-05-17 20:14 UTC	INCONCLUSIVE	Hypercontractive Term-Support Stable Rank Bounds Monotone Clique
2026-05-17 20:26 UTC	INCONCLUSIVE	S-Transform Free-Mult Defect Lower-Bounds DISJ Communication
2026-05-17 20:47 UTC	INCONCLUSIVE	Fekete Capacity of Laplacian Spectrum Upper-Bounds Max-Cut SoS-2
2026-05-17 21:20 UTC	INCONCLUSIVE	Polymer Cluster-Expansion Convergence Radius Bounds Tree-Res Size
2026-05-17 21:53 UTC	INCONCLUSIVE	Fatgraph Genus of Canonical Clause Ribbon Lower-Bounds Resolution
2026-05-17 22:25 UTC	FALSIFIED	Forman-Ricci Min-Curvature of Term-Overlap Graph Lower-Bounds Mon
2026-05-17 22:43 UTC	INCONCLUSIVE	Bourgain-Tzafriri Sub-Column Spectral Excess Bounds DISJ CC_R
2026-05-17 23:19 UTC	INCONCLUSIVE	Trivial-Isotype Mass Gap Separates Perm from Padded Det
2026-05-17 23:46 UTC	INCONCLUSIVE	Eulerian Descent Entropy Separates Perm from Padded Det_m
2026-05-18 00:33 UTC	INCONCLUSIVE	Signed-Laplacian Holonomy Spectrum Bounds Tseitin Tree-Resolution
2026-05-18 03:29 UTC	INCONCLUSIVE	Bonami-Beckner 4-2 Saturation of Clause-Sum Bounds DPLL Depth

Time	Verdict	Title
2026-05-18 03:58 UTC	INCONCLUSIVE	Cycle-Type Mixing Defect of Barrington BP Bounds NC^1 Length
2026-05-18 04:25 UTC	INCONCLUSIVE	NW-Design Restriction Dimension Separates Perm from Padded Det
2026-05-18 04:47 UTC	INCONCLUSIVE	Coxeter Inversion Spread of Barrington BPs Bounds NC^1 Depth
2026-05-18 05:16 UTC	INCONCLUSIVE	Birkhoff Spectral Gap of Gate Co-occurrence Bounds AC^0 PARITY Co

How to read the reports

- reports/supported_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified_findings.md — FALSIFIED with counterexamples.
- reports/daily_YYYY-MM-DD.md — chronological log by day.
- pvsnp_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

PDF export

Install pandoc + a LaTeX engine once (`sudo apt install pandoc texlive-xetex`), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
    pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```